Face Mask Detection with Live Alert System

1. Introduction

The project focuses on detecting whether people are wearing face masks in real-time. With the ongoing need for health safety, automated detection helps in monitoring public spaces and enforcing mask compliance efficiently. The system uses computer vision and deep learning techniques to identify masked and unmasked faces.

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2. Abstract

This project implements a face mask detection system using a Convolutional Neural Network (CNN). A dataset of masked and unmasked faces is used to train the model. The trained model is then integrated with OpenCV to process live images or sample images, detecting faces and identifying mask usage. Alerts are generated for unmasked faces to simulate real-time monitoring.

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3. Tools Used

- **Python 3.x**
- **TensorFlow / Keras** — for building and training CNN
- **OpenCV** — for image processing and face detection
- **Haar Cascades** — for detecting faces in images
- **Google Colab** — for training and demo
- **Matplotlib** — for displaying images in Colab

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4. Steps Involved in Building the Project

1. **Dataset Collection**
    - Used Kaggle Face Mask Detection dataset with two categories: with_mask and without_mask.
2. **Data Preprocessing**

- Resized images to 100x100 pixels
- Normalized pixel values
- Converted labels to one-hot encoding

3. **Model Training**
    - Built a CNN model using Keras Sequential API
    - Trained for 5 epochs
    - Achieved over 95% accuracy on the training set
4. **Simulated Real-Time Detection**
    - Loaded trained model in Colab
    - Used OpenCV to read sample images
    - Detected faces using Haar Cascades
    - Labeled images as Mask or No Mask
    - Displayed labeled images for demo purposes
5. **Alerts**
    - For unmasked faces, printed “ALERT: No Mask Detected!”
    - Simulated real-time monitoring

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5. Conclusion

This project demonstrates an effective approach to detecting face masks using deep learning and computer vision. The system can identify masked and unmasked faces with high accuracy and simulate alerts for unmasked individuals. It can be further extended to work with live webcams or security cameras for real-time surveillance. The approach is scalable and can be adapted to other object detection tasks.